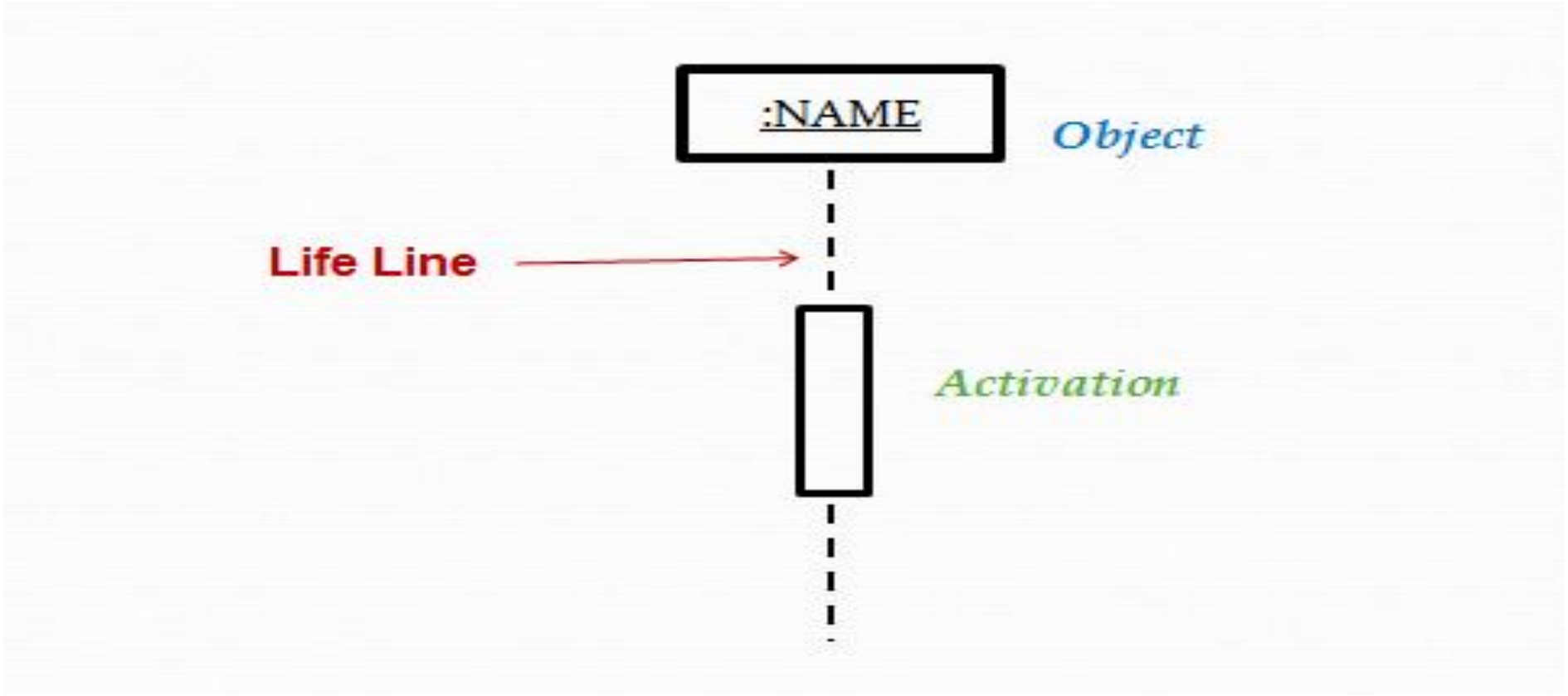
Sequence Diagram

Sequence Diagram

- Precise description of activities.
- Suitable as means for design and implementing communications.

- The **sequence diagram consists of**
  - **Objects** represented in the usual way (as named rectangles with the name underlined)
  - **Messages** represented as solid-line arrows
  - **Time** represented as a vertical progression.

2-Sequence Diagram



3- Messages

- There are two types of messages

1. **Call (synchronous) message**

   - This is a request from the object sending the message to another object that receives the message.
   - Usually, this entails the **sender waiting for the receiver** to carry out that operation.





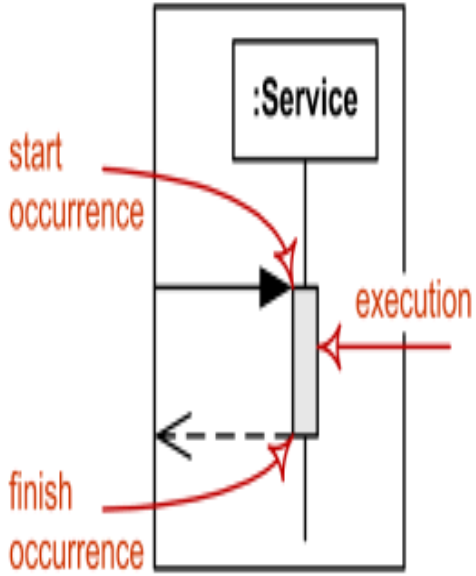
2. **Asynchronous message**

   Sender **transfers control to the receiver** and **doesn't wait for the operation to complete.**



4-Time

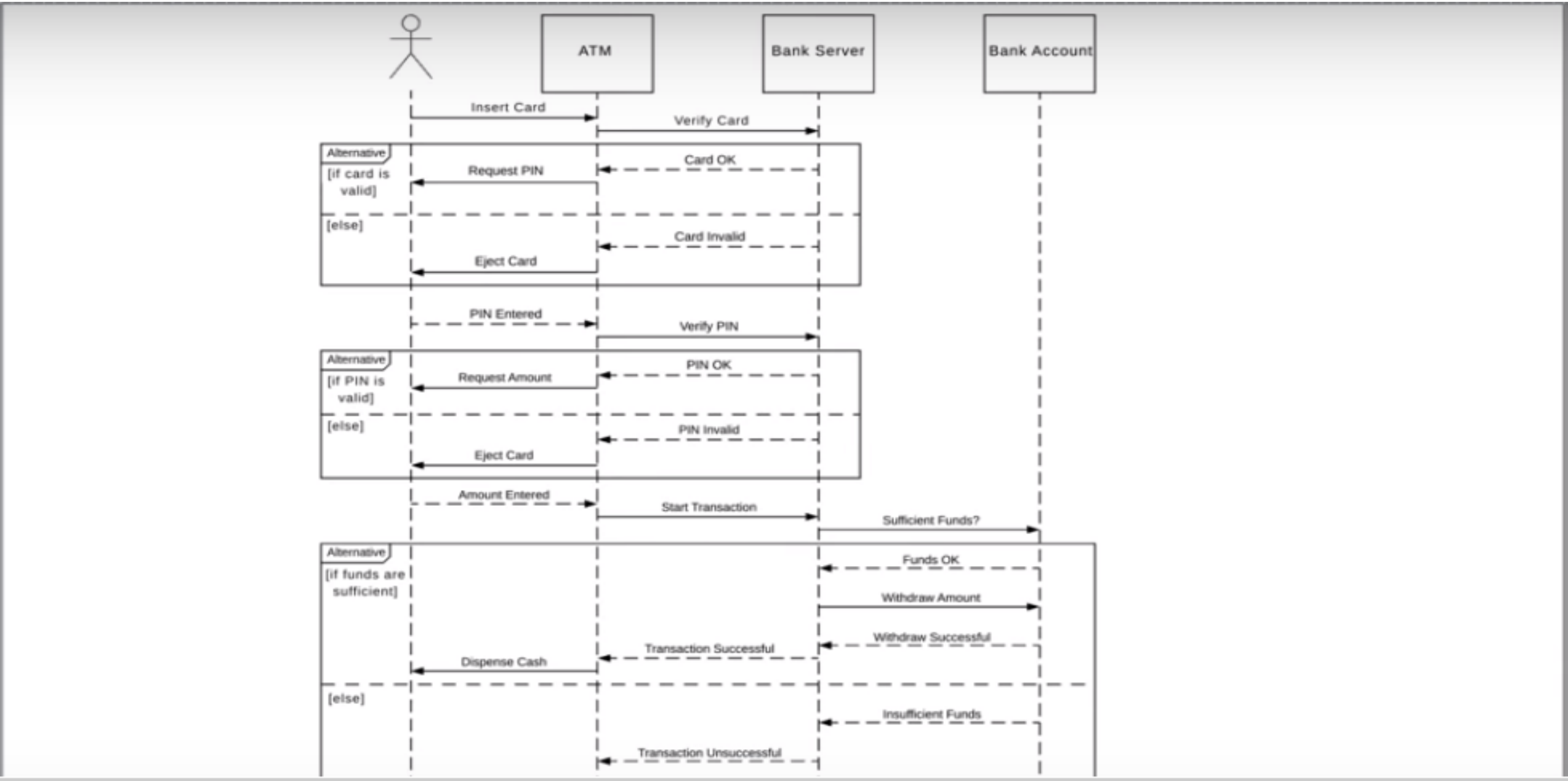
- **Execution** (full name - **execution specification**, informally called **activation**) is **interaction fragment** which represents a period in the participant's lifetime when it is
- executing a unit of behavior or action within the **lifeline**,
- sending a signal to another participant,
- waiting for a reply message from another participant.
- The **duration** of an execution is represented by two
- **execution occurrences** - the **start** occurrence and the
- **finish** occurrence.
- **Execution** is represented as a thin grey or white rectangle
- on the lifeline.



Duration of an execution is represented by two execution occurrences - start and finish.

Sequence for ATM

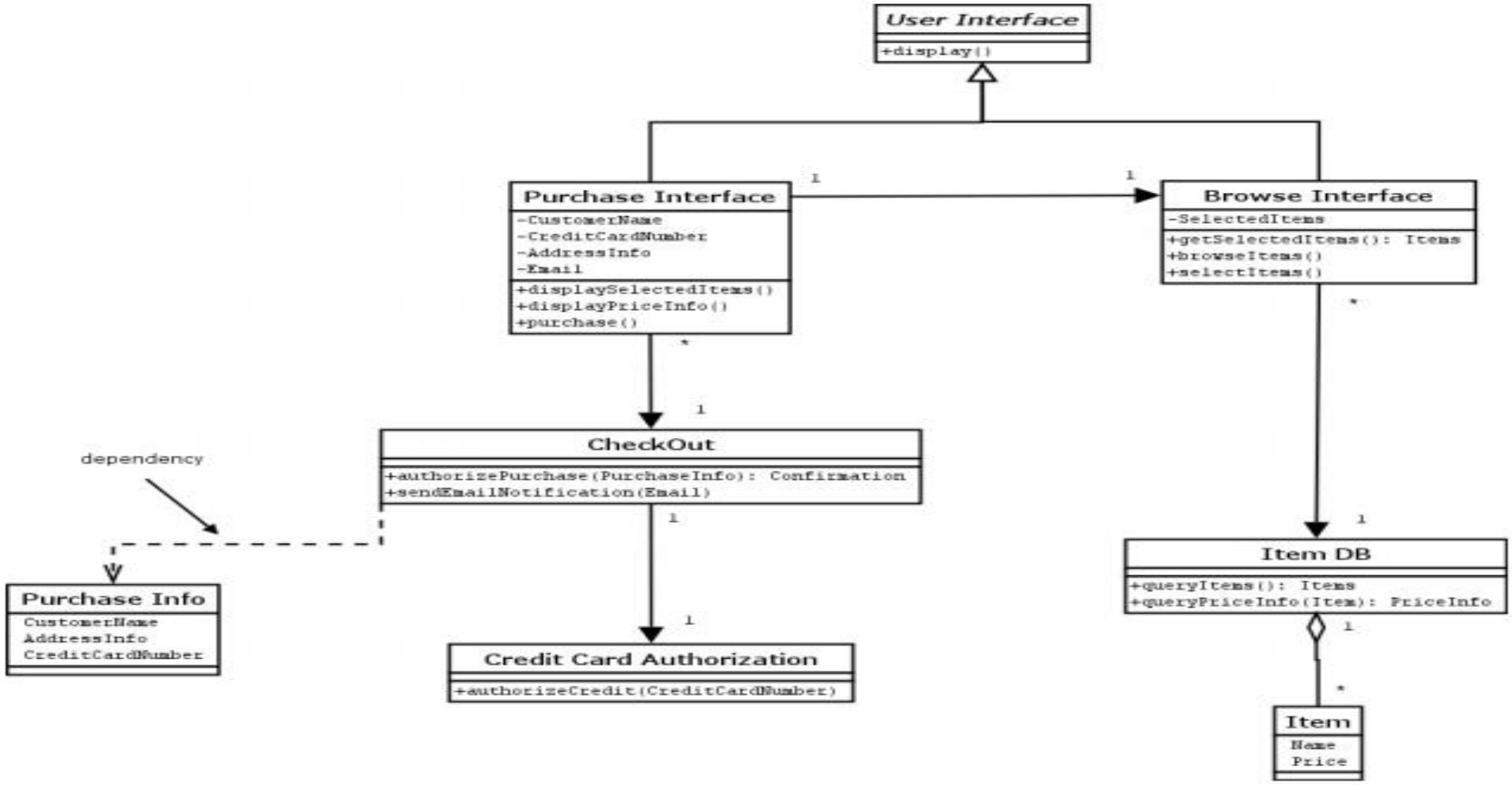
Sequence for ATM



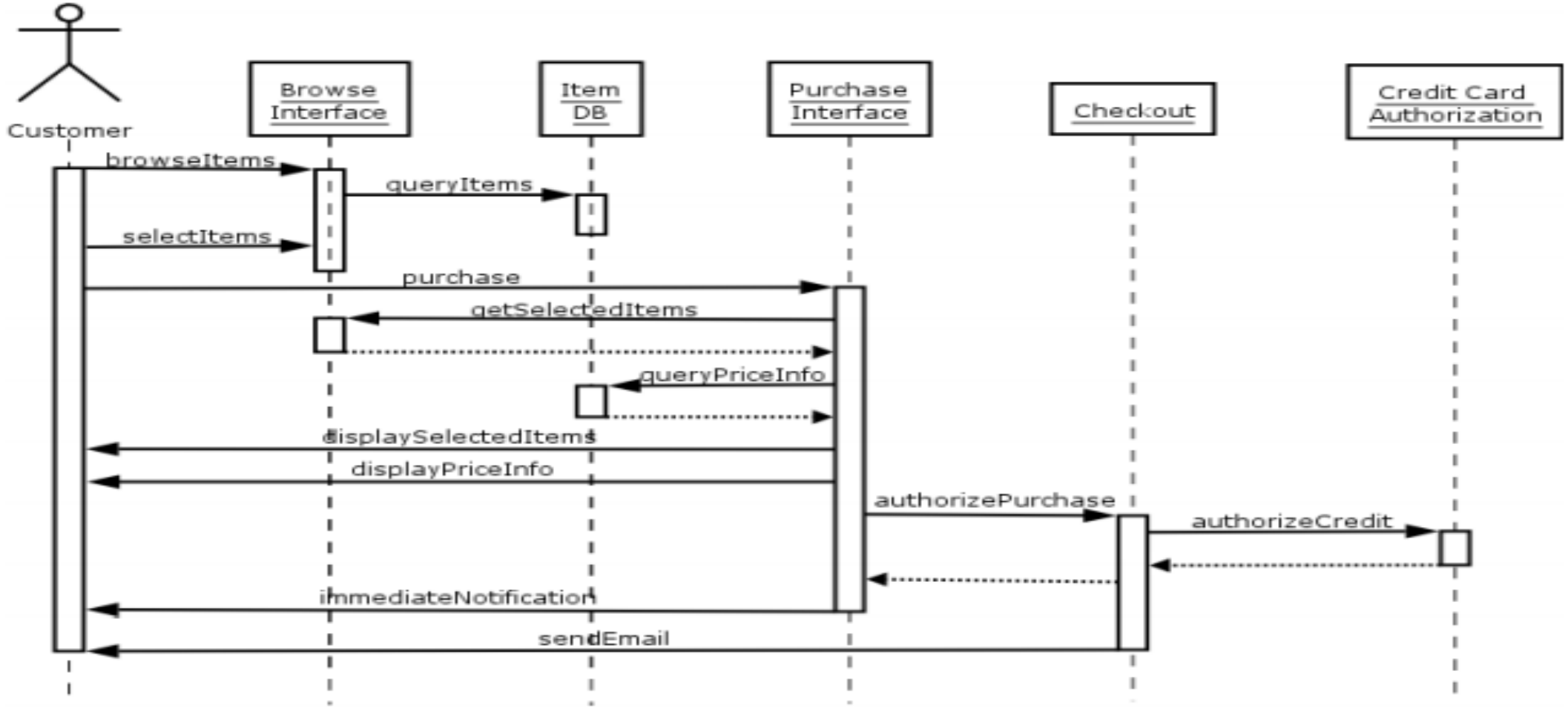
Example: Online Shopping

- In order to develop our sequence diagram we will need the use case text and the specification class diagram: Use Case Text: Buy a Product Online
  - 1. Customer browses through catalog and selects items to buy.
  - 2. Customer goes to checkout.
  - 3. Customer fills out shipping information.
  - 4. System presents full pricing information, including shipping information.
  - 5. Customer fills in credit card information.
  - 6. System authorizes purchase.
  - 7. System confirms sale immediately.
  - 8. System sends confirming email to customer. The specification class diagram as developed previously:

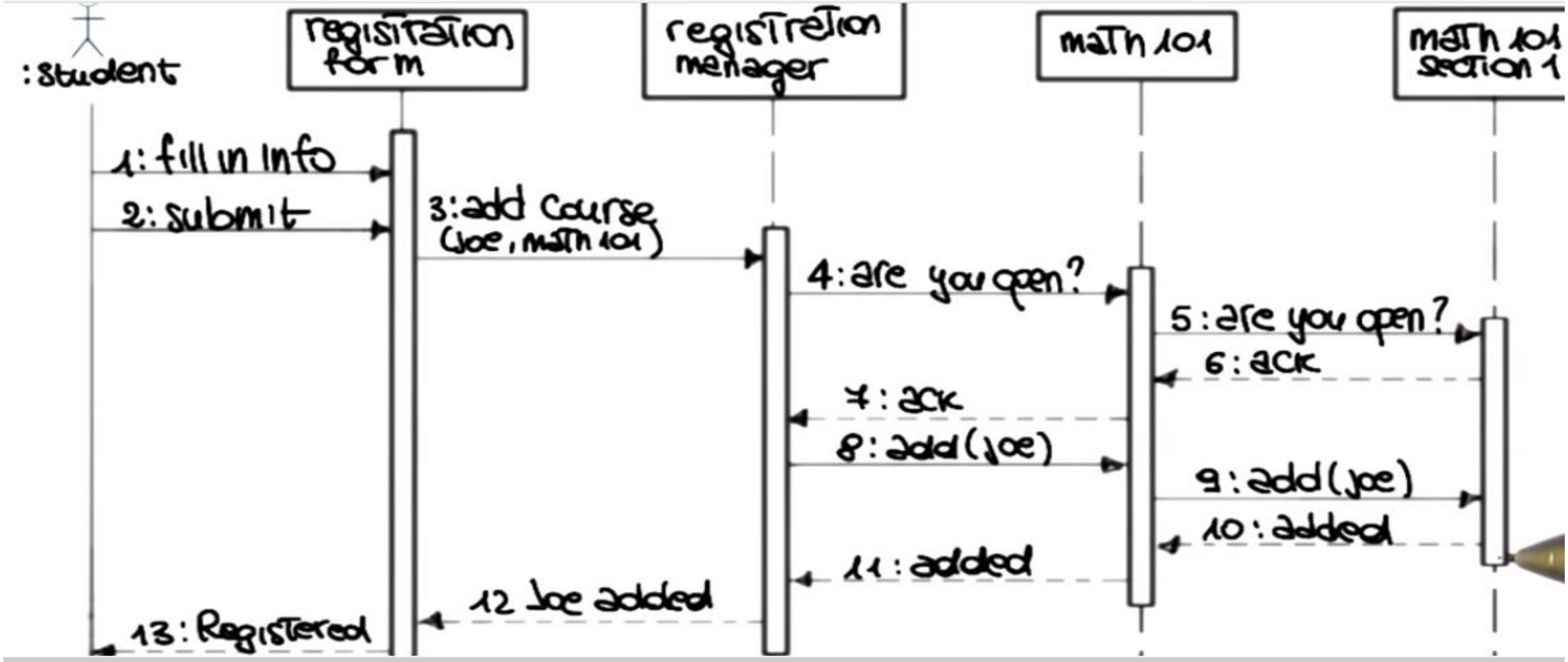
Class Diagram



Sequence Diagram for online shopping



Sequence Diagram for Registration Form



Thank you